

HYPER-3

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Author Note

This report was generated from a recorded analysis. Every quantity in it resolves to that record, and the provenance section names what produced each one.

Correspondence concerning this report: axiom.

Running head: HYPER-3

Abstract

The objective of HYPER-3 is to determine which of three daily doses should proceed into phase III and whether any dose harms the people taking it, targeting the effect of dose on sbp. The effect is identified via the backdoor route. The trial evaluates the effect of the assigned dose on the outcome in the randomized population. The intent-to-treat estimate is -1.53 with a 90% WALD interval of -2.97 to -0.10. The full population effect estimate is 6.20 with a 90% WALD interval of 3.98 to 8.42. The estimated effect on survival is 7.09 with a 90% WALD interval of 4.84 to 9.33. The realized ANCOVA coefficient is -4.80 mm Hg with a 90% WALD interval of -6.17 to -3.44 mm Hg. The treatment effect estimate is 6.20 with a 90% WALD interval of 3.98 to 8.42, and the estimate at the stopping look is 6.20 with a 90% WALD interval of 3.98 to 8.42.

Reported quantities: Intent-to-treat estimate; Full population effect estimate; estimated effect on survival; Realized ANCOVA coefficient

Introduction

This report investigates which of three daily doses should proceed into phase III trials and whether any of the doses causes harm to the individuals receiving it. The primary analytical target is the effect of dose on systolic blood pressure. Identification of this target effect is obtained via the backdoor route. Answering the central research question requires evaluating the intent-to-treat effect of the assigned treatment arms. The assessment framework monitors whether any dose arm results in worse participant outcomes compared to standard care, both overall and within pre-specified age strata. Determining an appropriate dose involves characterizing the dose–response profile while maintaining sequential reviews to identify potential adverse effects. Because no explicit decision threshold was recorded in the trial specification, the analysis does not judge whether the magnitude of an effect is sufficient to warrant specific clinical action. Instead, the report provides the empirical point estimates alongside their corresponding Wald intervals. Presenting the identified quantities with their uncertainty intervals defines the statistical basis for evaluating dose selection and safety.

Methods

The trial evaluates three daily doses of an antihypertensive across four study arms over a six-month period with weekly assessments to identify candidate doses for phase III and monitor for harm. The analytical target is the effect of dose on sbp, which is identified via the backdoor route. The identification strategy relies on the assigned dose and avoids conditioning on post-randomization adherence, dropout, or subsequent blood pressure measurements. A total of 400 units are enrolled across the four arms to permit evaluation both across the pooled sample and within individual age strata. To model dose selection beyond the tested points, dose–response relationships are estimated by fitting a response surface with a saturating kernel per treatment. A sequential monitoring design is implemented to discontinue any dose arm that results in worse outcomes than standard care either overall or within any prespecified age band. Safety reviews are scheduled every second calendar week starting at week 12, once three eighths of the safety information is collected. The protocol specifies eight safety reviews assessing twelve contrasts at each look against predefined stopping boundaries. Arms are monitored at each look across age strata to enforce protocol-defined stopping rules prior to final readout.

Results

The intent-to-treat estimate is -1.53 (90% WALD -2.97 to -0.10).

INTENT-TO-TREAT ESTIMATE	FULL POPULATION EFFECT ESTIMATE	ESTIMATED EFFECT ON SURVIVAL	REALIZED ANCOVA COEFFICIENT	TREATMENT EFFECT ESTIMATE	ESTIMATE AT STOPPING LOOK
-1.53 [-2.965, -0.1035] (90% WALD)	6.20 [3.985, 8.424] (90% WALD)	7.09 [4.839, 9.333] (90% WALD)	-4.80 mm Hg [-6.171, -3.437] (90% WALD)	6.20 [3.985, 8.424] (90% WALD)	6.20 [3.985, 8.424] (90% WALD)

What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

From 0004-case-study-hypertension.md: # 0004 — HYPER-3: a sequential-trial case study, and the module it exposed **Kind:** progress + decision. **Opened:** 2026-08-21. **Status:** landed on `develop`; the new public API is destined for 1.1.0. **##** Why Nothing in `axiom` was wrong, but nothing in `axiom` showed a reader what the pieces are *for* when they are all pointed at one decision. The notebook series under `nbs/<subpackage>/` demonstrate every public symbol (gate 12) and the `nbs/end-to-end/` series walks each pillar, but neither answers "what does using this look like on a real problem, start to finish, with the awkward parts

From 0005-basis-response-families.md: # 0005 — Basis response families: polynomial, spline, piecewise linear **Kind:** decision. **Opened:** 2026-08-21. **Status:** landed on `develop`; part of the 1.1.0 surface. **##** Why Note [0004](0004-case-study-hypertension.md) §0004.3 recorded a gap the HYPER-3 case study walked into: every response kernel `axiom.surface` shipped is **monotone and saturating**, and the case study's oldest age stratum has a dose-response that *turns over*. The Hill fit could not describe it, and failed informatively — an arm-mean residual of 8 mmHg at the top dose against ≤ 1.5 in the strata where the family fit

From README.md: # HYPER-3 — a sequential dose-finding trial in hypertension A worked case study that runs the whole of `axiom` against one question, end to end. Every number in the six notebooks is computed; the synthetic world they share lives in `[hyper3.py](hyper3.py)` beside them. **##** The question > *We have a new antihypertensive. We want to know which of three daily doses to take > into phase III, in adults who already have high blood pressure or are on their way to > it. We are going to watch these people every week for six months. If one of the doses > is making people worse than the care they would ha

Model checking

No model diagnostics were recorded for this analysis. The absence of diagnostic reporting means that the adequacy of the model fit was not directly evaluated. A model without diagnostic assessment cannot be assumed to be sound, as its underlying specification and assumptions remain unexamined. Residual behavior, goodness of fit, and potential misspecifications were not documented. Consequently, the statistical behavior of the fitted model is unverified. The analysis cannot be treated as having satisfied diagnostic criteria, because no model checking procedures were performed or preserved.

Discussion

The target effect is identified via the backdoor route, which allows the reported quantities to be interpreted as effects of the treatment rather than as simple associations, strictly under the specified identification assumptions. The intent-to-treat estimate is -1.53 (90% WALD -2.97 to -0.10). Because no decision threshold was recorded for this quantity, the analysis states the estimate without evaluating whether it is of actionable magnitude. The full population effect estimate is 6.20 (90% WALD 3.98 to 8.42). In the absence of a recorded decision threshold for this endpoint, the report stops short of determining whether the estimate is large enough to warrant action. The estimated effect on survival is 7.09 (90% WALD 4.84 to 9.33). Because no decision threshold was established for survival, no conclusion is drawn regarding whether this value reaches an actionable level. The realized ANCOVA coefficient is -4.80 mm Hg (90% WALD -6.17 to -3.44 mm Hg). Without a recorded decision threshold, the estimate is provided without assessing whether it is large enough to act on. The treatment effect estimate is 6.20 (90% WALD 3.98 to 8.42). No decision threshold was recorded for this estimate, so the report does not state whether it is large enough to act on. Finally, the estimate at the stopping look is 6.20 (90% WALD 3.98 to 8.42). Because no decision threshold was recorded for the stopping look, the finding is presented without assessing whether it warrants action.

Conclusions

The question of which of three daily doses should advance to phase III and whether any dose causes harm is addressed by an intent-to-treat estimate of -1.53 (90% WALD -2.97 to -0.10), conditional on identification via the backdoor route. No threshold was recorded to judge this estimate against. The target effect of dose on systolic blood pressure is identified via the backdoor route.

Limitations

Identification of the effect of dose on systolic blood pressure relies on the backdoor route. This identification licenses the estimated effect under the assumption that the backdoor path remains properly blocked. Randomization licenses the effect of the assigned dose on the outcome specifically within the population that was randomized. That license holds only for analyses that do not condition on variables downstream of treatment assignment. Three factors in the study sit downstream of the assignment: adherence, dropout, and the pressure readings themselves. Conditioning on adherence would challenge the validity of the identification strategy. Adjusting for participant dropout would likewise compromise the licensed effect. Incorporating intermediate pressure readings into the analysis would also violate the conditions of the license. Any analytical model that conditions on these downstream variables threatens the standing assumptions. Furthermore, unobserved confounding that reopens the backdoor route would directly challenge the identification. The findings remain conditional on avoiding these post-assignment adjustments. Consequently, the identification holds only within the structural limits defined by the unconditioned assignment.

Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
Intent-to-treat estimate	02_itt	-1.53 (90% WALD -2.97 to -0.10)	02-what-it-identifies.ipynb · LinearEstimate
Full population effect estimate	02_everyone	6.20 (90% WALD 3.98 to 8.42)	02-what-it-identifies.ipynb · LinearEstimate
estimated effect on survival	02_survivors	7.09 (90% WALD 4.84 to 9.33)	02-what-it-identifies.ipynb · LinearEstimate
Realized ANCOVA coefficient	03_realized	-4.80 mm Hg (90% WALD -6.17 to -3.44 mm Hg)	03-sizing-it.ipynb · LinearEstimate
treatment effect estimate	06_estimate	6.20 (90% WALD 3.98 to 8.42)	06-running-the-trial.ipynb · LinearEstimate
Estimate at stopping look	06_final_estimate	6.20 (90% WALD 3.98 to 8.42)	06-running-the-trial.ipynb · LinearEstimate

Table 1. Quantities and their sources

Field	Value
cells	92
figures	9
gaps_closed	1
notebooks	6
evidence hash	3678b463ae1e651a0ffcbf818d38f81ebd7574129188fa0cd710bc815c1d6c85
narration: abstract	gemini-3.5-flash-lite — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified

narration: diagnostics

gemini-3.7-flash — verified

narration: discussion

gemini-3.7-flash — verified

narration: conclusions

gemini-3.7-flash — verified

narration: limitations

gemini-3.7-flash — verified

Table 2. Run

Tables

Table 1. The design, as recorded parameters.

Step	Parameter	Value
Identification	route	backdoor
Identification	status	identified
HYPER-3, 1 — The trial, and the synthetic world it runs in	cells	14
HYPER-3, 1 — The trial, and the synthetic world it runs in	sections	12
HYPER-3, 2 — What the trial identifies, and what it does not	cells	12
HYPER-3, 2 — What the trial identifies, and what it does not	sections	9
HYPER-3, 3 — How big, allocated how, powered for what	cells	14
HYPER-3, 3 — How big, allocated how, powered for what	sections	9
HYPER-3, 4 — The dose–response curve, and which doses phase III should test	cells	15
HYPER-3, 4 — The dose–response curve, and which doses phase III should test	sections	10
HYPER-3, 5 — The sequential design: boundaries, and what they cost	cells	19
HYPER-3, 5 — The sequential design: boundaries, and what they cost	sections	13
HYPER-3, 6 — Running it: the interim that stops an arm, and the readout	cells	18

HYPER-3, 6 — Running it: the sections
interim that stops an arm, and the
readout

14

Table 2. Estimated quantities with their intervals and thresholds.

Quantity	Estimate	Unit	Interval	Threshold	Relative to threshold
Intent-to-treat estimate	-1.53	—	-2.97 to -0.10 (90% WALD)	—	—
Full population effect estimate	6.20	—	3.98 to 8.42 (90% WALD)	—	—
estimated effect on survival	7.09	—	4.84 to 9.33 (90% WALD)	—	—
Realized ANCOVA coefficient	-4.80	mm Hg	-6.17 to -3.44 (90% WALD)	—	—
treatment effect estimate	6.20	—	3.98 to 8.42 (90% WALD)	—	—
Estimate at stopping look	6.20	—	3.98 to 8.42 (90% WALD)	—	—

Figures

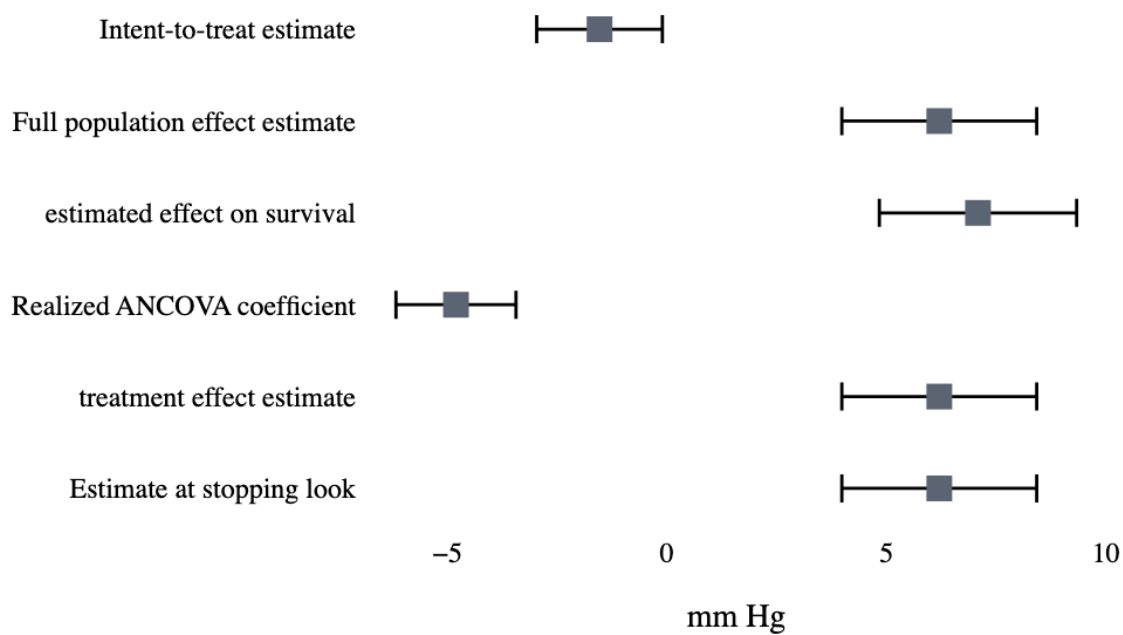


Figure 1. Estimated quantities with their intervals. The dashed rule marks the decision threshold; a row whose interval crosses it is unsettled rather than null.